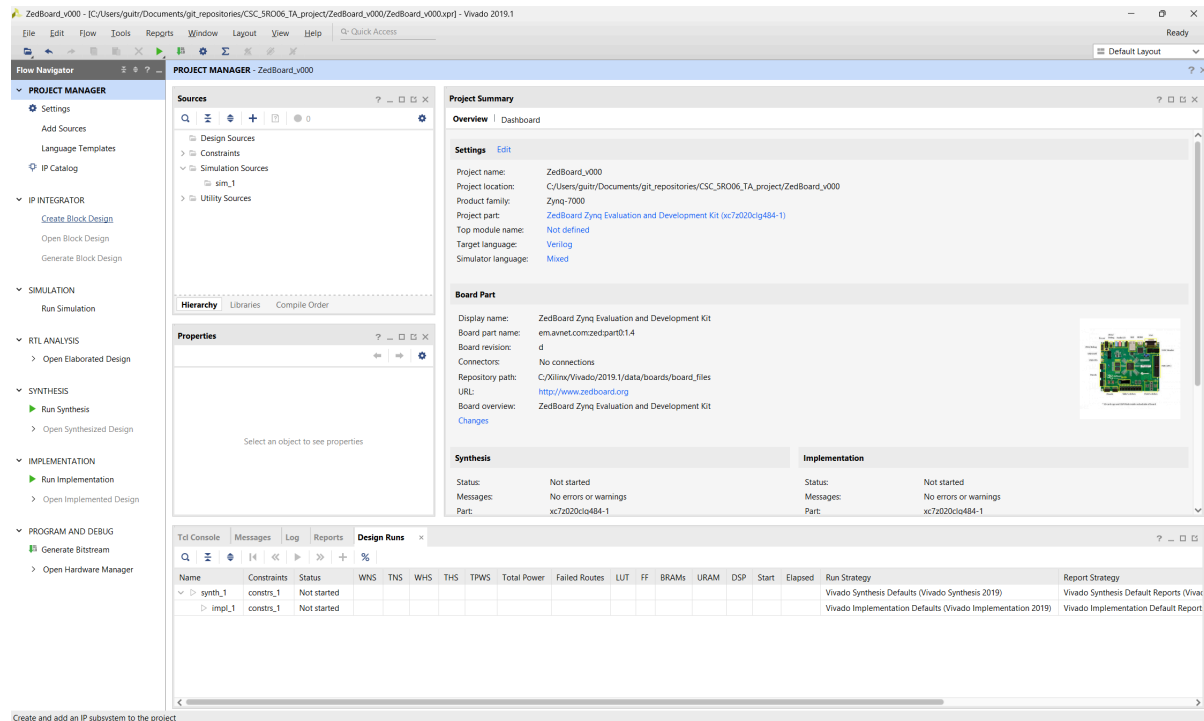


Vivado 2019.1 - Create Block Diagram

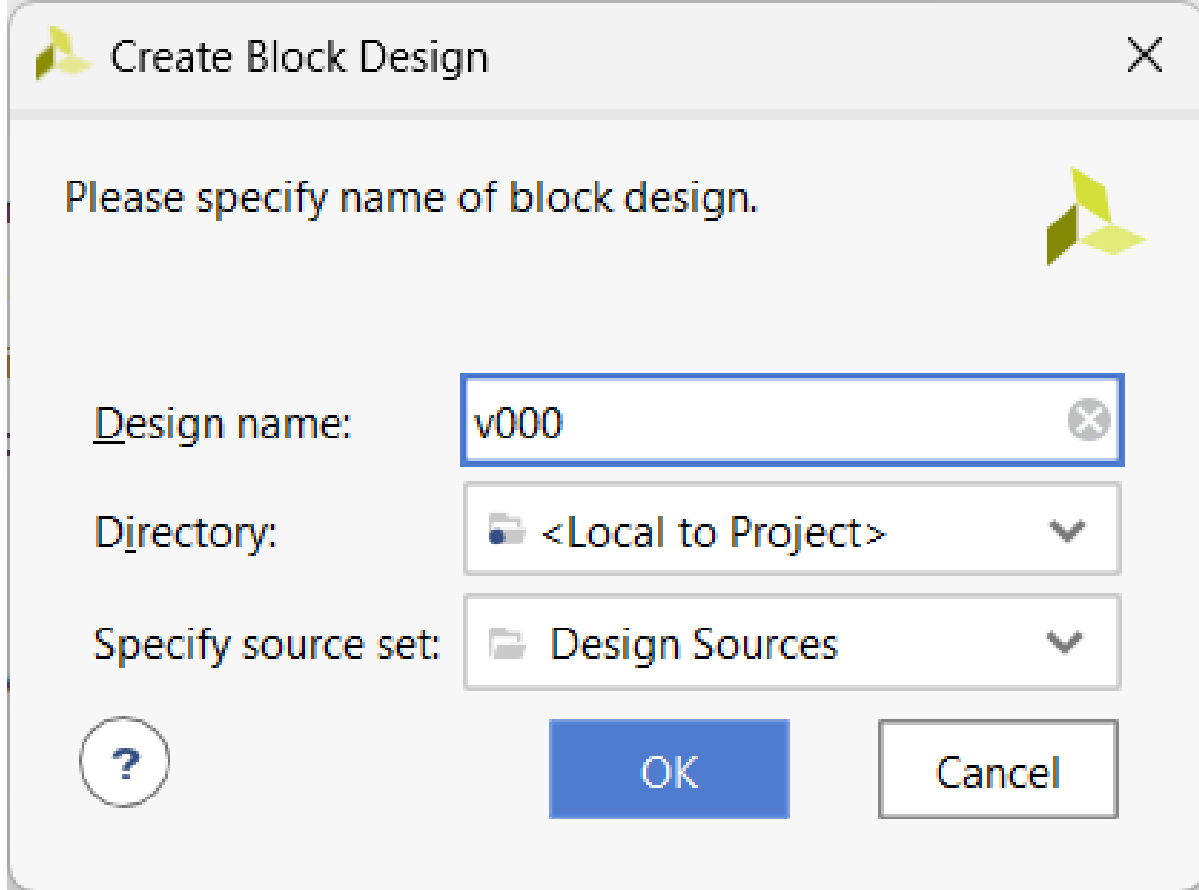
Block Diagram

IP Integrator

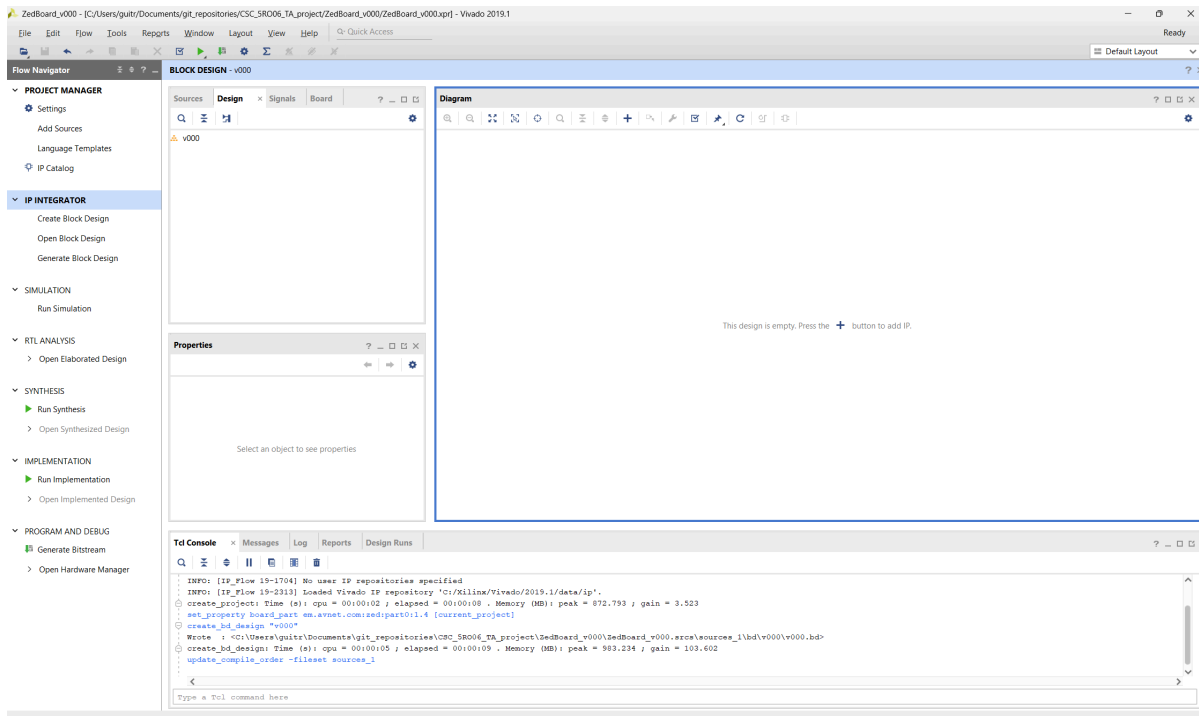
In IP INTEGRATOR click in Create Block Design.



In Create Block Design name your Block Diagram and **do not modify** other configurations.

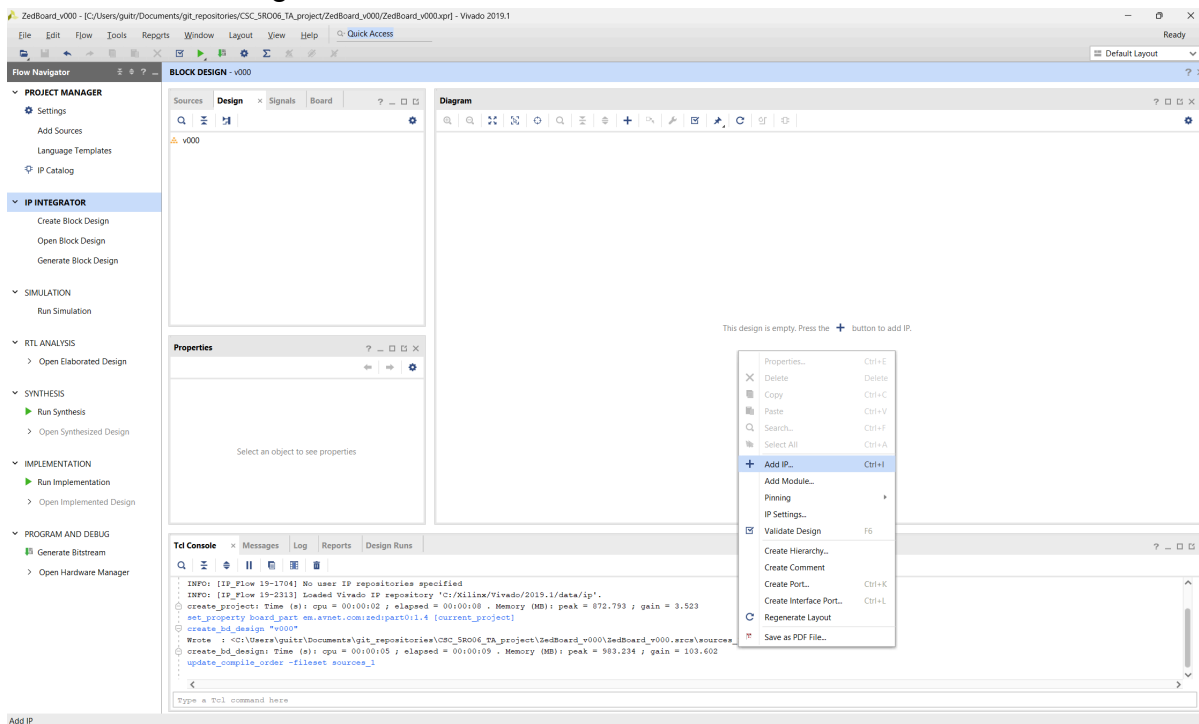


File will be created and is ready for use. Vivado interface will change as presented:



Add IP

Add device with Right Click and Add IP

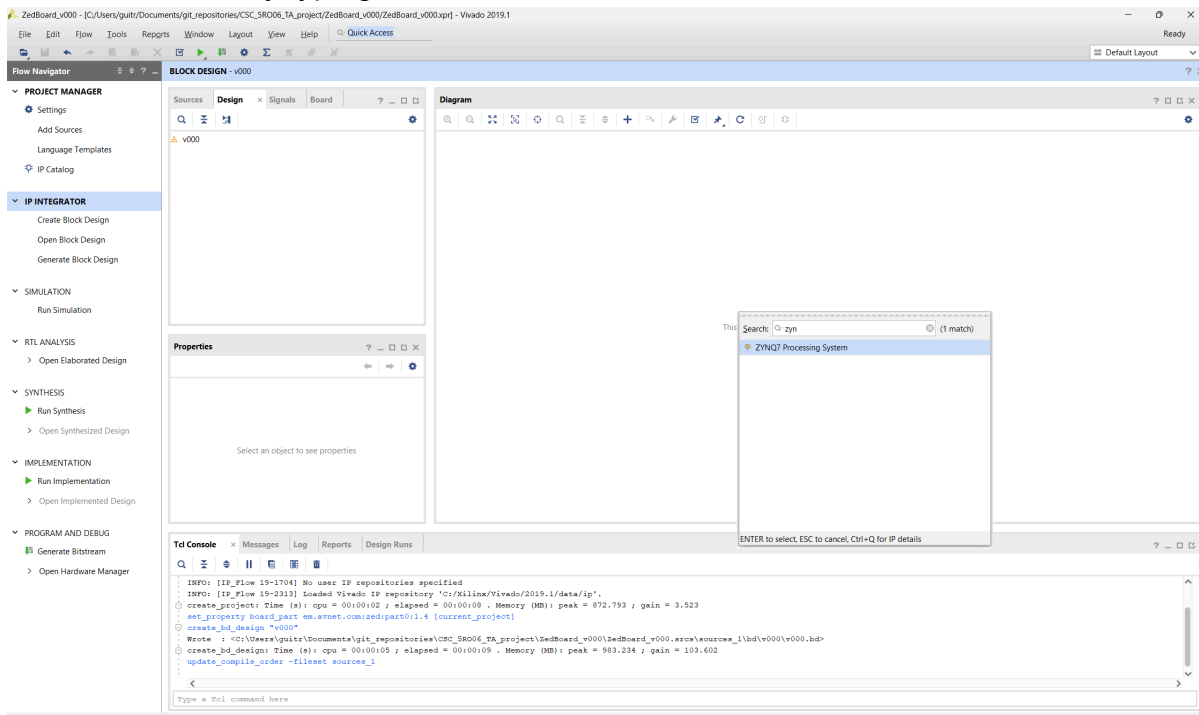


Observation

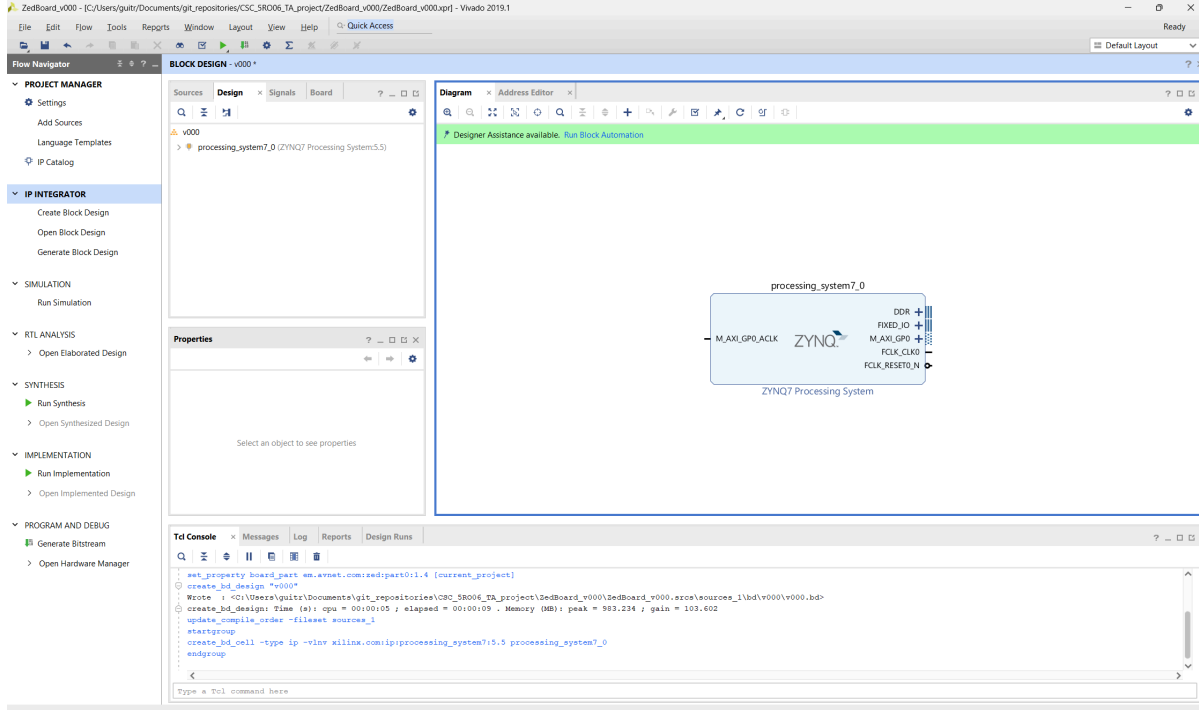
Intellectual Property is an IP in the Vivado. Refers to pre-designed and pre-verified modules that you can use in your FPGA designs.

Zynq

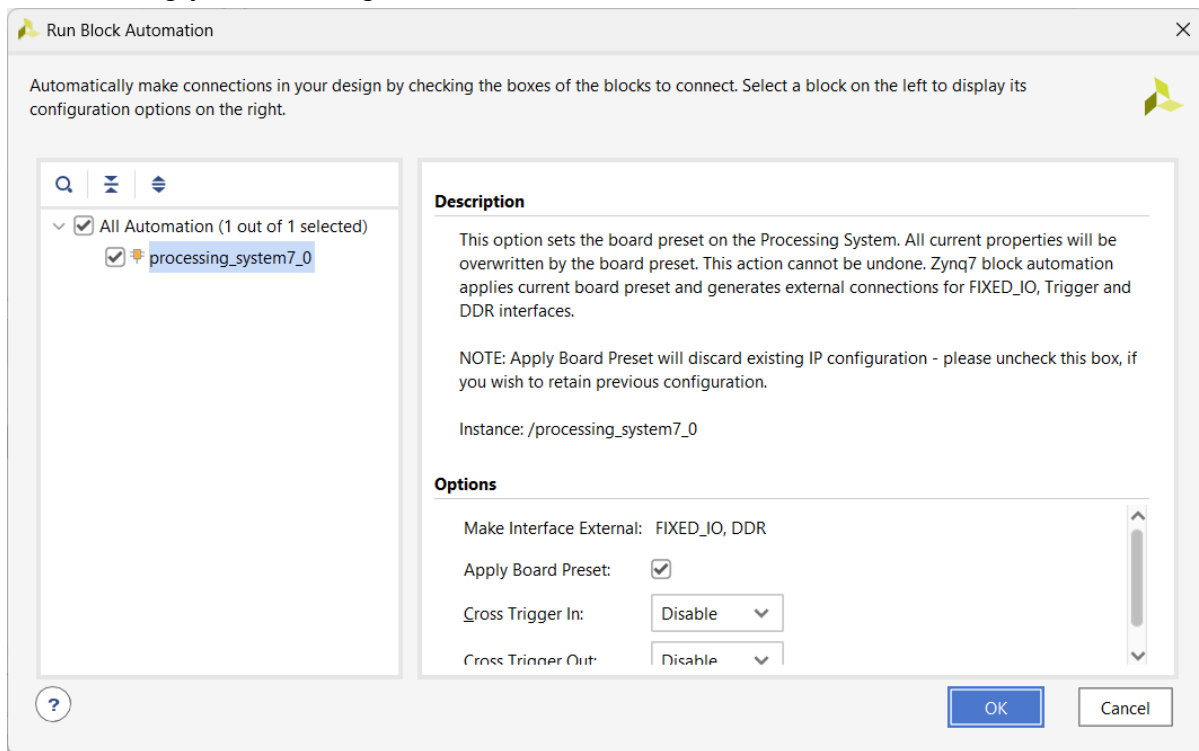
Add main FPGA by typing the ZYNQ code.



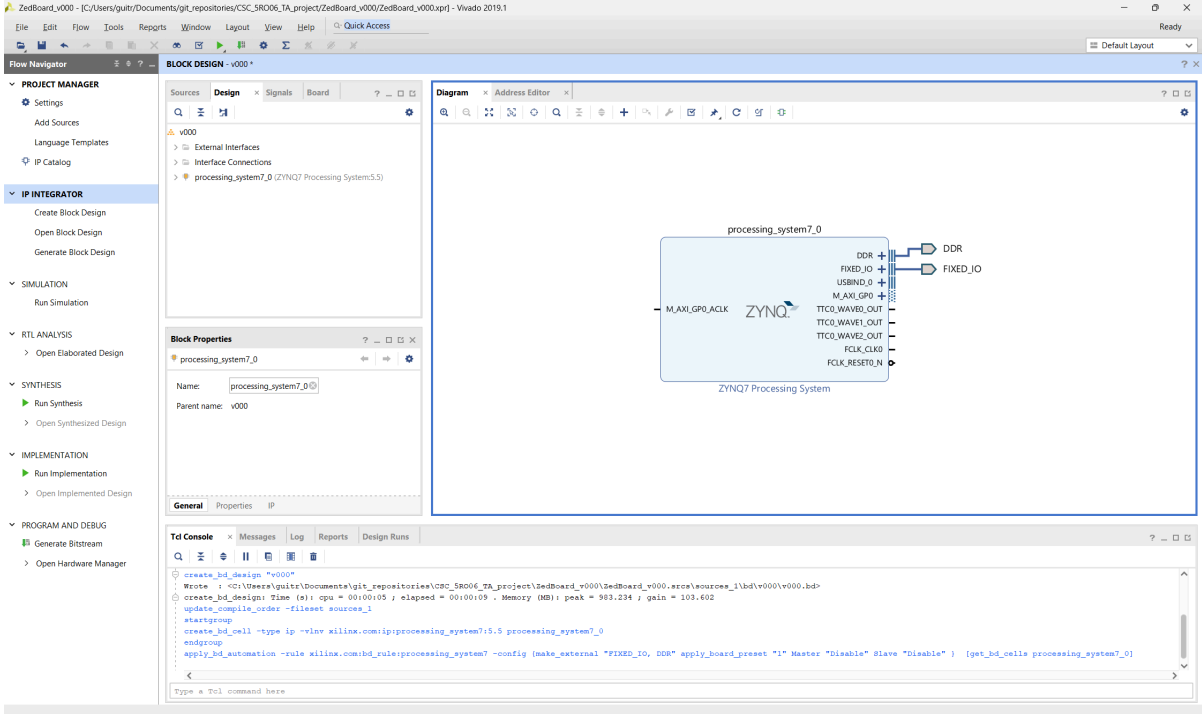
After included, a Device will appear in the Block Diagram window.



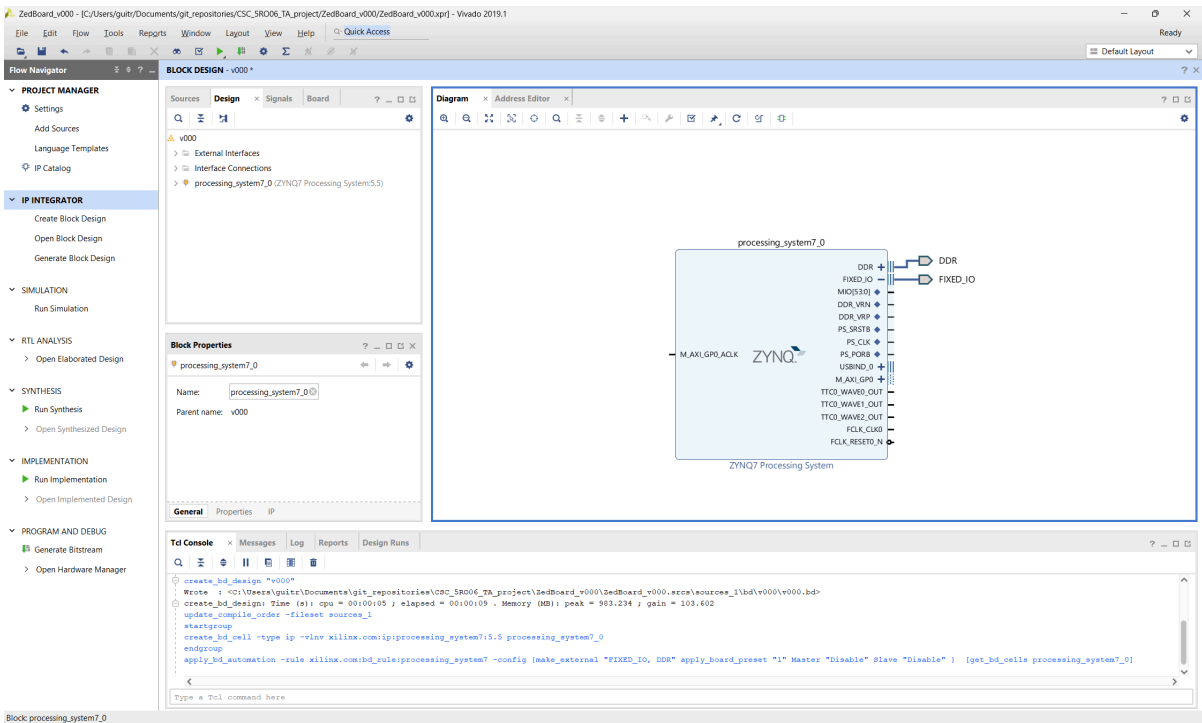
After adding you block, right click and "Run Block Automation"



Default configuration is used. The DDR3 memory is necessary for the FPGA behavior.

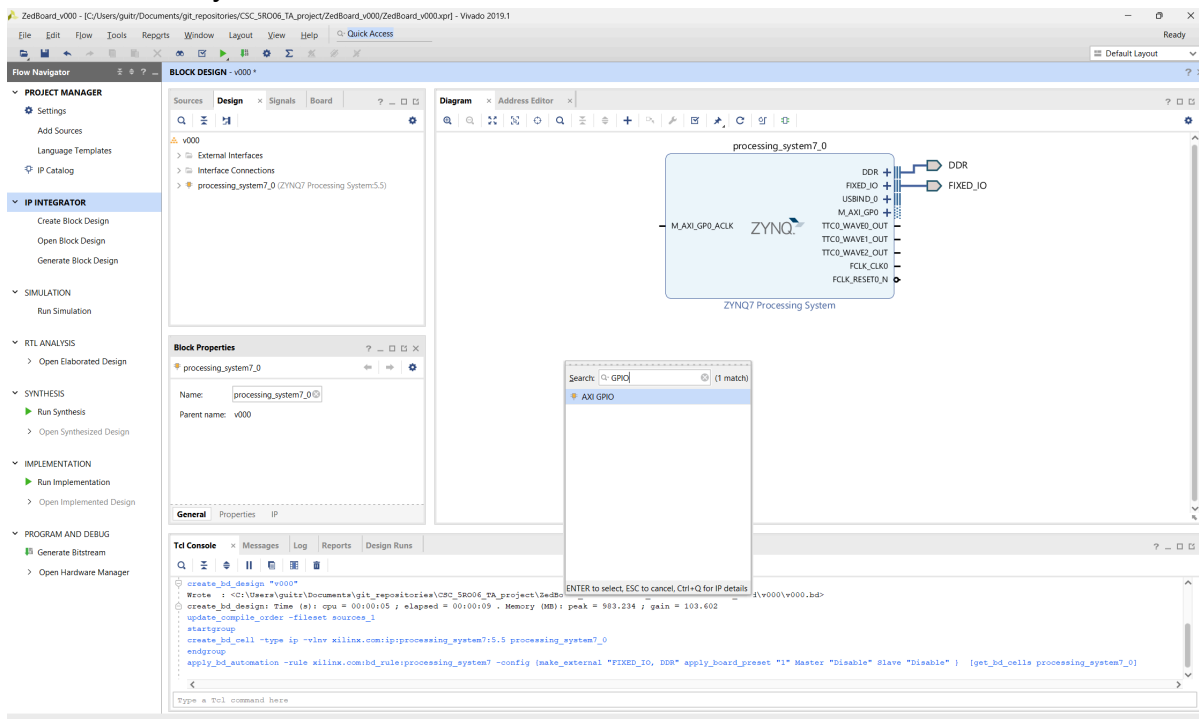


Each pin may be expanded to show all its connections as shown below:

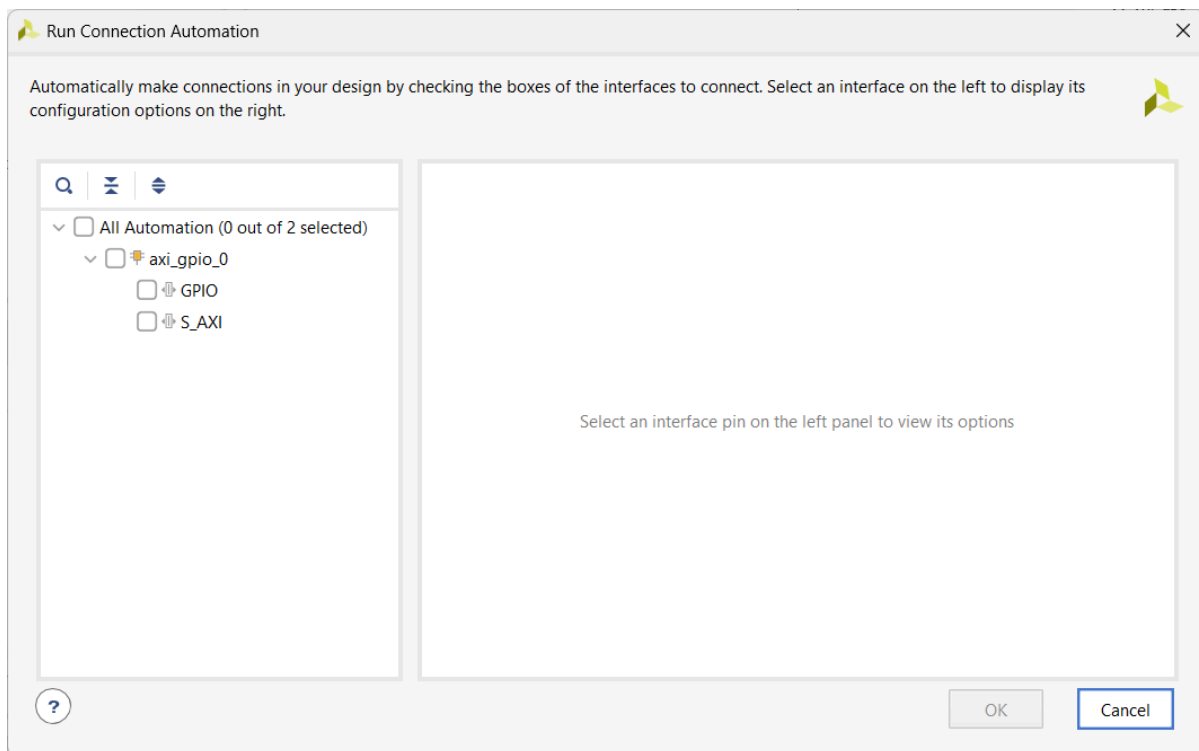


GPIO

Another IP maybe included such as GPIO:



Run Block Automation for the GPIO block.



Check all automations for the GPIO block.

Validate Design

Create and Package New IP...
Create Interface Definition...
Enable Partial Reconfiguration...
Run Tcl Script...
Property Editor
Associate EIE Files...
Generate Memory Configuration File...
Compile Simulation Libraries...
Download Latest Boards...
Xilinx [d] Store...
Custom Commands
Language Templates
Settings...

Block Interface Properties

Name: GPIO
Mode: MASTER
Connection: axi_gpio_1_GPIO
Associated clock: None

General Properties

Tcl Console

```
INFO: [board_rule 100-100] set_property CONFIG.GPIO_BOARD_INTERFACE leds_bits [get_bd_cells /axi_gpio_1]  
INFO: [board_rule 100-100] create_bd_intf_port -mode Master -vlnv xilinx.com:interface:gpio_rtl:1.0 leds_bits  
INFO: [board_rule 100-100] connect_bd_intf_net /leds_bits /axi_gpio_1/GPIO  
endgroup  
set_property location {1 224 386} [get_bd_cells axi_gpio_1]  
set_property location {3 308 -115} [get_bd_cells rst_ps7_0_100M]  
set_property location {3 333 116} [get_bd_cells rst_ps7_0_100M]  
set_property location {1 273 393} [get_bd_cells axi_gpio_1]
```

Validate Design

Validation successful. There are no errors or critical warnings in this design.

OK

If you did by hand maybe there will be problems. If you had errors check [Vivado HLS 2019.1 - Troubleshooting](#)